

Sinusoid

Montague Hamilton

2 October 2026

1 Questions

1.

$$h(t) = c + a \sin\left(\frac{2\pi t}{p} + \phi\right)$$

For a wheel centred at height 4, radius 3, period 12; the rider starts at the top. At $t = 3$,

$$\begin{aligned} h(t) &= 4 + 3 \sin\left(\frac{2\pi 3}{12} + \frac{\pi}{2}\right) \\ &= 4 + 3 \sin(\pi) \\ &= 4 \end{aligned}$$

So at the same level as the wheels centre.

2. Describe fully (amplitude, midline, period): $y = 5 \sin(2x) - 1$

The midline is -1 , so the function starts at -1 and centres around it.

The amplitude is 5, so the function reaches 5 higher and 5 lower than midline. So the maximum would be 4 and the minimum would be -6 .

The period is 2, so the function repeats twice for every increase in x .

3. Convert a period of 6 seconds – what is the angular frequency? And what phase ϕ describes a rider who starts at the midline and moves downwards?

Frequency is the inverse of the period, so

$$\begin{aligned} f &= \frac{1}{p} \\ f &= \frac{1}{6} \end{aligned}$$

The two simplest phases that work for a rider starting at the midline and moving downwards is $\pm \frac{\pi}{2}$